

Research Proposal

Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

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Thesis: The capstone is now deliberately pruned. It studies one clinically interpretable workflow: LLM-drafted pharmacogenomic or genomic medication-alert text. The evaluation asks whether a pre-display evidence gate reduces overclaiming relative to guideline-supported evidence without inappropriately denying bounded, aligned alerts.

Problem Statement

Medical LLMs can sound authoritative even when the evidence only supports a narrower statement. In pharmacogenomics this matters because a real guideline, a verified citation, a population-frequency result, or a variant assertion may support a medication-safety note without proving broad patient-outcome benefit, universal population transport, or deterministic action.

Hypothesis

Compared with ungated LLM alert text, a three-gate controller should reduce evidence overclaiming while tracking inappropriate denial in a future independent evaluation. The supported Stage 1 claim is narrower: the implementation follows its intended action grammar on author-designed synthetic archetypes, while also exposing two primary-gate precedence mismatches.

Methods

- Workflow: pharmacogenomic/genomic medication-alert text, including medication-gene pairs and variant assertions relevant to prescribing or safety review.
- Gates: endpoint/actionability, population fit, and citation/guideline support.
- Stage 1: 30 author-designed synthetic cases with expected decisions; executable controller reports action conformance, primary-gate conformance, designed overclaim archetypes blocked, bounded alerts allowed, inappropriate denial, and gate-precedence error categories. Stage 1 consumes structured evidence annotations only; free-text extraction is out of scope.
- Stage 2: replace author-designed cases with independently authored cases and blinded reviewer adjudication by qualified clinicians or pharmacogenomics reviewers.

Stage 1 Progress

Metric	Current result	Interpretation
Cases	30	Synthetic case bank for the pruned workflow.
Author-designed overclaim share	20/30	Stress-test coverage, not a measured clinical base rate.
Overclaim archetypes allowed unchanged	0/20	Derived from overclaim cases that reached allow unchanged.
Action / primary-gate conformance	30/30; 28/30	PGX19 and PGX24 are gate-precedence mismatches.
Inappropriate denial	0	Count of bounded aligned alerts blocked too strongly.

Value

The contribution is a submission-ready narrowing move: instead of presenting a universal assurance theory, the paper proposes a small auditable benchmark for a recurrent clinical workflow. This directly answers the critique that the prior package was broad without a single causal chain.

Selected Sources

- ML4H 2026 home. <https://ml4h.ahli.cc/>
- ML4H 2026 call for participation. <https://ml4h.ahli.cc/submit/call-for-papers/>
- Byrne, D. W., Domenico, H. J., & Moore, R. P. (2024). Artificial intelligence for improved patient outcomes: The pragmatic randomized controlled trial is the secret sauce.. <https://kjonline.org/DOIx.php?id=10.3348%2Fkjr.2023.1016>
- Stegenga, J. (2018). Medical Nihilism. Oxford University Press.. <https://global.oup.com/academic/product/medical-nihilism-9780198747048>
- FDA-NIH BEST Resource. <https://www.ncbi.nlm.nih.gov/books/NBK326791/>
- CPIC guidelines overview. <https://cpicpgx.org/guidelines/>
- ClinPGx CPIC CYP2C19-clopidogrel guideline. <https://www.clinpgx.org/guideline/PA166251443>
- CPIC CYP2C19-clopidogrel 2022 update. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9287492/>
- CPIC DPYD-fluoropyrimidines guideline. <https://www.clinpgx.org/guideline/PA166251462>
- CPIC DPYD-fluoropyrimidines update. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5760397/>
- CPIC TPMT/NUDT15-thiopurines update. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12997511/>
- ClinPGx HLA-B-abacavir guideline. <https://www.clinpgx.org/guideline/PA166251444>
- ClinPGx HLA-A/HLA-B-carbamazepine guideline. <https://www.clinpgx.org/guideline/PA166251448>
- ClinVar. <https://www.ncbi.nlm.nih.gov/clinvar/>
- gnomAD. <https://gnomad.broadinstitute.org/>
- All of Us Research Program. <https://allofus.nih.gov/>
- Genome India project. <https://genomeindia.in/>
- Psifas / Mosaic. <https://partnership.psifas.org.il/>